Supplementary note: shortest-path multiplicity on the diamond hierarchical lattice

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A short, self-contained derivation of the number of shortest paths (geodesics) on the (b, s) diamond hierarchical lattice, using only the recursive construction. The count between the two poles is obtained in closed form, $P_t = b^{(s^t-1)/(s-1)}$ (i.e. 2^{2^t-1} for the diamond), and the average over all vertex pairs is shown to grow doubly exponentially, $\bar{\sigma}_t \sim \kappa\, 2^{2^t}/4^t \approx 2.1\, P_t/n_t$ with $\kappa \approx 1.576$. All formulas are verified against direct construction of the lattice (`multiplicity.py`). This note is for internal inspection; it is not yet part of the manuscript.

I. SETUP: THE DEFLATION RECURSION

We use the same (b, s) diamond hierarchical lattice as the main text, grown from a single bond between the poles A, B by replacing every edge with b parallel paths of s edges. It is convenient here to read the construction *backwards* (deflation): G_t consists of b parallel branches between A and B, and each branch is s copies of G_{t-1} placed in series and joined at new hub vertices. For the standard diamond $b = s = 2$ this is particularly simple: G_t is two branches, each two copies of G_{t-1} joined at a middle hub, so G_t is four copies of G_{t-1} glued on the corner 4-cycle A–M_1–B–M_2, where M_1, M_2 are the two generation-1 hubs and each side of the cycle is one copy G_{t-1} with its poles at the two adjacent corners.

Because distinct copies meet only at these corner vertices, any path leaving a copy must exit through one of its two corner poles. Two consequences are used repeatedly:

(i) *Copies are isometric subgraphs.* Leaving a copy and returning costs at least a full renormalized bond, so for two vertices of the same copy the graph distance equals the in-copy distance, and every geodesic between them stays inside the copy.

(ii) *Corner-transit.* A geodesic between vertices of *different* copies is forced to pass through corner vertices, so it factorizes into an in-copy piece, a corner-to-corner piece on the 4-cycle, and a second in-copy piece.

Throughout, $\sigma(u, v)$ denotes the number of shortest paths between u and v, $d(u, v)$ their distance, D_t the pole-to-pole distance, and $P_t \equiv \sigma(A, B)$.

II. EXTREMAL MULTIPLICITY: THE TWO POLES

Across one series copy, lengths add and geodesic counts multiply; across the b parallel branches (all of the same length) counts add. Hence

$$D_t = s\, D_{t-1} = s^t, \qquad P_t = b\, P_{t-1}^s, \qquad P_0 = 1. \tag{1}$$

Taking $\log_b$ of the recursion gives $L_t \equiv \log_b P_t = 1 + s\, L_{t-1}$ with $L_0 = 0$, so $L_t = (s^t - 1)/(s - 1)$ and

$$\boxed{P_t = b^{(s^t-1)/(s-1)}} \qquad \Longrightarrow \qquad P_t = 2^{2^t-1} \ \ (b = s = 2). \tag{2}$$

For the diamond this also reads $P_t = 2^{D_t-1}$ with $D_t = 2^t$: the number of geodesics between the antipodal poles is *doubly exponential* in the generation. Table I lists the exact values; Eq. (2) was checked against direct construction for $(b, s) \in \{(2, 2), (3, 2), (2, 3), (4, 2), (3, 3)\}$.

III. AN ADDITIVITY LEMMA

Lemma. Every vertex lies on a pole-to-pole geodesic: $d(v, A) + d(v, B) = D_t$ for all $v \in G_t$.

Proof (induction on t). True for G_0 (a single edge). Assume it for G_{t-1}. A vertex v lies in some copy, say the copy between corners X and Y; by the inductive hypothesis $d(v, X) + d(v, Y) = D_{t-1} = s^{t-1}$. One of X, Y is the near pole side and the other leads on to the far pole through $s - 1$ further copies in series, each of length D_{t-1}; since detours

TABLE I. Pole-to-pole distance $D_t = 2^t$ and geodesic multiplicity $P_t = 2^{\,2^t-1}$ for the diamond ($b = s = 2$), verified against direct construction.

t	$D_t = 2^t$	$P_t = 2^{\,2^t-1}$
1	2	2
2	4	8
3	8	128
4	16	32 768
5	32	2 147 483 648
6	64	9 223 372 036 854 775 808
7	128	$\approx 1.70 \times 10^{38}$

around the parallel branches are strictly longer, the distance to the far pole is the in-copy distance to the appropriate corner plus $(s-1)D_{t-1}$. Adding the two pole distances gives $s\,D_{t-1} = s^t = D_t$. □

Two corollaries are used below. First, the distance profile of a vertex is fixed by a single number $\alpha(v) \equiv d(v, A)$, since $d(v, B) = D_t - \alpha(v)$. Second, adjacent corners of the 4-cycle are at distance D_{t-1} with P_{t-1} geodesics between them, while opposite corners (A–B and M_1–M_2) are at distance D_t with P_t geodesics; that is, the corner 4-cycle is a renormalized copy of G_1 with each edge carrying length D_{t-1} and multiplicity P_{t-1}.

IV. AVERAGE OVER ALL PAIRS

Let $S_t = \sum_{u<v} \sigma(u, v)$ be the total geodesic count and $\bar{\sigma}_t = S_t/\binom{n_t}{2}$ the average multiplicity, with $n_t = \frac{1}{3}(2 \cdot 4^t + 4)$ and $n_t = 4n_{t-1} - 4$. The deflation splits pairs into those inside a single copy and those spanning two copies:

$$S_t = 4\,S_{t-1} + \mathrm{Inter}_t. \tag{3}$$

Intra-copy pairs contribute $4S_{t-1}$ because copies are isometric [§I(i)] and each intra-copy pair, including each adjacent-corner pair, belongs to exactly one copy.

The inter-copy term is a corner-transit convolution [§I(ii)]. Parameterize u by $p \equiv$ its distance to a reference corner of its copy, with a_u, b_u the numbers of shortest paths from u to the two corners; likewise r, a_v, b_v for v. For u, v in *opposite* copies (e.g. the A–M_1 and B–M_2 copies), evaluating the four corner routes with the lemma gives the compact result

$$d(u, v) = 2^t - |p - r|, \tag{4}$$

$$\sigma(u, v) = P_{t-1}\Big[\,a_u b_v\,\mathbf{1}_{p>r} + b_u a_v\,\mathbf{1}_{p<r} + (a_u b_v + b_u a_v)\,\mathbf{1}_{p=r}\Big]. \tag{5}$$

The factor P_{t-1} is the corner-to-corner traversal; the bracket convolves the two copies' pole-path counts, selected by which exit corner is shortest. Adjacent copies (sharing one corner) give an analogous, slightly longer, case list. Equations (4)–(5) make explicit why the sum is dominated by *near-antipodal* pairs (small $|p - r|$): those inherit a geodesic count of order $P_{t-1}^2 \sim P_t$, comparable to the pole value.

Carrying out (3) exactly yields Table II. Numerically the average obeys the doubly-exponential law

$$\bar{\sigma}_t \;\sim\; \kappa\,\frac{2^{\,2^t}}{4^t} \;\approx\; 2.1\,\frac{P_t}{n_t}, \qquad \kappa \to 1.576, \tag{6}$$

i.e. $\bar{\sigma}_t = \Theta(P_t/n_t)$: the average tracks the *pole* multiplicity divided by the vertex count, up to an $O(1)$ prefactor κ. The prefactor is the fixed point of the corner-transit convolution and is not elementary; Eq. (6) with $\kappa \approx 1.576$ is the honest closed characterization. (The two forms are equivalent since $2^{2^t} = 2P_t$ and $n_t \to \frac{2}{3}4^t$.)

V. REMARK: THE SEAM EIGENVALUE AGAIN

The whole effect is another face of the seam eigenvalue b of the main text. A geodesic threads $\sim \log_s(\text{distance})$ successive scales, and at each scale it independently chooses one of the b parallel branches; the branch choices multiply,

TABLE II. Total geodesic count $S_t = \sum_{u<v} \sigma(u,v)$ and average multiplicity $\bar{\sigma}_t = S_t/\binom{n_t}{2}$ for the diamond, from the recursion (3), checked against direct construction. The last two columns show the convergence of Eq. (6): $\kappa = \bar{\sigma}_t\, 4^t/2^{2^t}$ and $\bar{\sigma}_t\, n_t/P_t$.

t	n_t	S_t	$\bar{\sigma}_t$	κ	$\bar{\sigma}_t\, n_t/P_t$
1	4	8	1.333	1.333	2.667
2	12	128	1.939	1.939	2.909
3	44	6 912	7.307	1.827	2.512
4	172	6 148 096	418.1	1.633	2.194
5	684	1.556×10^{12}	6.663×10^{6}	1.589	2.122
6	2732	2.652×10^{22}	7.110×10^{15}	1.579	2.106
7	10924	—	3.274×10^{34}	1.576	2.102

giving $b^{\#\text{scales}}$ alternatives. Along a full pole-to-pole geodesic the number of scales is $(s^t-1)/(s-1)$, reproducing $P_t = b^{(s^t-1)/(s-1)}$ of Eq. (2). The same branch-choice degeneracy, thinned by how close a generic pair is to antipodal, produces the average (6). In the language of the RG map M, the exponential growth of the interface $e_{12} = b^t$ (the seam eigenvalue) and the exponential growth of the geodesic count share the same origin: the b-fold parallel branching of the cell.

VI. REPRODUCIBILITY

All numbers above are produced by `multiplicity.py`, which builds G_t by edge replacement and computes shortest-path counts by breadth-first search with exact (arbitrary-precision) integer accumulation of σ. It prints Table I (pole distance and multiplicity, and the general-(b,s) check) and Table II (total and average multiplicity, with the asymptotic ratios).